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# Technology-Mediated Scaffolding to Support Critical Thinking in Pre-Service Teachers' Academic Writing

**Silvana Arriagada Anabalón *, Miguel Nussbaum Voehl, Nicolás Navarro and Joaquín Araya**

Department of Computer Science, Faculty of Engineering, Pontificia Universidad Católica de Chile, Santiago 6904411, Chile; mn@uc.cl (M.N.V.); nbnavarro@uc.cl (N.N.); jiaraya4@uc.cl (J.A.)
* Correspondence: sbarriagada@uc.cl

**Abstract**

Developing critical thinking skills (CTS) is a central goal of higher education, yet pre-service teachers often receive little explicit guidance on how to apply them in academic or professional writing. This study reports the design, implementation, and initial examination of IDEA, a purpose-built platform for technology-mediated scaffolding in academic writing. The scaffold targets four CTS subskills: interpretation (clarifying meaning), analysis (examining relationships among ideas), explanation (articulating and justifying reasoning), and evaluation (judging claims against criteria). Using a design-based research (DBR) approach, 57 fourth-year pre-service language teachers at a highly selective Chilean university completed three structured writing tasks during one semester. The first two tasks were completed without IDEA; the third used the full scaffold. Students' texts were assessed with an expert-reviewed analytical rubric. Cohen's kappa for independent pre-consensus ratings was 0.88 for interpretation, 0.79 for analysis, 0.84 for explanation, and 0.83 for evaluation; Cronbach's alpha across the pooled 171 text-level observations was 0.790, reported as a descriptive indicator of score coherence. Omnibus repeated-measures ANOVAs were statistically significant for all four subskills, and 11 of the 12 Bonferroni-adjusted pairwise comparisons were significant. Because the study had no control group and IDEA use coincided with the final task, these changes cannot be attributed to IDEA on the basis of this design. Focus group participants ($n$ = 12) reported that the scaffold made CTS more concrete, helped them organize disciplinary reasoning, and increased their awareness of planning, self-evaluation, and revision. These findings concern critical-thinking performance in writing; overall writing quality was not measured separately.

**Keywords:** critical thinking skills; teacher education; technology-mediated scaffolding; pre-service teachers; instructional design

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## 1. Introduction

Critical thinking skills (CTS) are a central objective of higher education and are widely recognized as essential for 21st-century professional practice (Jaramillo et al., 2025; Nuryana et al., 2024). They are especially important in teacher education because future teachers must use critical thinking in their own professional decisions and cultivate it in their students (Marangio et al., 2024; Romero Ariza et al., 2024). CTS encompass interrelated cognitive operations—including interpretation, analysis, explanation, and evaluation—that enable sound reasoning, evidence-based judgment, and reflective practice (Facione, 1990, 2000; Halpern, 1998; Behar-Horenstein & Niu, 2011). Yet institutional mission statements

often invoke critical thinking without explaining how students should apply it, and pre-service teachers report limited guidance on using these skills in academic and professional contexts (Wason, 2025; Yuan et al., 2022; H. Zhang et al., 2020).

Academic writing provides an important context for enacting and practicing CTS because it requires students to organize arguments, evaluate sources, explain their reasoning, and draw inferences (Fraser et al., 2024). When structured as an epistemic activity, writing can make reasoning available for reflection and revision (Fraser et al., 2024; Ghanizadeh, 2017; Wingate, 2012). Without systematic support, however, students often rely on descriptive summary, integrate evidence weakly, or present unclear argumentative structures (Fraser et al., 2024; Stapleton, 2011). These difficulties are particularly consequential in teacher education, where a single written text may require students to integrate pedagogical theory, disciplinary knowledge, and classroom-based reasoning.

A further challenge is that writing assignments rarely specify the reasoning required by each critical thinking skill. When pre-service teachers are asked to analyze a pedagogical experience, assess an instructional resource, or reflect on teaching practice, they must integrate disciplinary knowledge, pedagogical reasoning, and contextual interpretation. Evaluation, for example, requires students to establish criteria, justify judgments, and consider implications, yet assignments seldom make these expectations explicit. Pre-service teachers may therefore struggle when the epistemic demands of a task remain tacit or insufficiently defined (Lithoxoidou & Papadopoulou, 2024; Fraser et al., 2024). Scaffolding in authentic contexts can help connect abstract skill labels with situated reasoning (Wason, 2025); otherwise, critical thinking risks remaining a rhetorical expectation rather than a practiced competence.

Digital scaffolds can make complex reasoning steps visible during composition. Embedded prompts, examples, and formative feedback can guide learners through higher-order reasoning at key stages of the writing process (Benabbes et al., 2025; Eragamreddy & Joseph, 2025; Williams & Bower, 2025). In teacher education, these tools may support the written enactment of critical thinking while making the targeted reasoning operations more explicit.

Although previous studies have examined critical thinking in teacher education and digital support for writing, less is known about how a non-generative digital scaffold can translate particular critical-thinking subskills into visible writing actions in authentic teacher-education tasks. To address this gap, the present study examines IDEA, a purpose-built non-generative digital scaffold, focusing on critical-thinking performance in interpretation, analysis, explanation, and evaluation; overall writing quality was not treated as a separate outcome. Two research questions guided the study:

- RQ1: How did pre-service teachers' performance in the four subskills change across a fixed sequence of three academic writing tasks, the first two completed without IDEA and the third completed with it?
- RQ2: How did focus group participants perceive IDEA as supporting or constraining their enactment of these subskills while writing?

The study contributes an initial DBR-cycle account of a purpose-built platform designed to organize students' reasoning through structured prompts, self-evaluation, and checklist-triggered prewritten feedback aligned with the targeted CTS subskills, without generating or reformulating students' text.

## 2. Theoretical Framework

### *2.1. Conceptualizing and Operationalizing Critical Thinking Skills*

Defining critical thinking (CT) is essential for designing pedagogical support. Although theoretical perspectives differ, most describe CT as an overarching construct in-

volving interrelated cognitive skills used in reasoning, decision-making, problem-solving, argumentation, and reflective judgment (Facione, 1990; Halpern, 1998; Paul & Elder, 2006). Table 1 summarizes the four CT dimensions selected for this study, presenting theory-informed working definitions and illustrative writing actions that clarify their operationalization in the platform and rubric.

**Table 1.** Critical-thinking subskills: working definitions and instructional design descriptors (based on M. J. Davies & Willing, 2023; Facione, 1990; Indriani, 2019; Jamil et al., 2024; Susongko et al., 2024; Thomas & Lok, 2015; Tseng, 2020).

| Critical-Thinking Subskill | Working Definition | Illustrative Writing Actions |
|---|---|---|
| Interpretation | Clarifying and contextualizing meaning by categorizing information and determining its significance. | Identify main ideas; paraphrase; classify information; interpret patterns or data; clarify ambiguity. |
| Analysis | Examining the structure of arguments and the relationships among claims, reasons, evidence, and assumptions. | Compare viewpoints; decompose arguments; identify assumptions; trace logical connections; identify supporting evidence. |
| Explanation | Articulating reasoning coherently by presenting and justifying claims, procedures, or conclusions with relevant reasons or evidence and explicit criteria or considerations. | State claims, results, or conclusions; give reasons; explain relevant criteria, procedures, or methods; connect evidence to claims; make the reasoning explicit and coherent. |
| Evaluation | Judging the credibility, relevance, and logical strength of claims or arguments against explicit criteria. | Apply criteria; assess evidence and reasoning; identify inconsistency or bias; justify judgments; consider counterarguments. |

Interpretation, analysis, explanation, and evaluation were selected from Facione's (1990) Delphi framework because they could be aligned directly with the course's writing tasks and represented through observable textual evidence. These four skills have also been identified in previous technology-mediated work as central components through which critical thinking can be supported and enacted in learning contexts (Nussbaum et al., 2021). The selection was therefore theory-informed and task-aligned rather than a comparative claim that these are the four most validated or universally central CT skills. The dimensions are analytically distinguishable but can overlap in a single written passage (Behar-Horenstein & Niu, 2011; Thomas & Lok, 2015). The actions in Table 1 are synthesized instructional design descriptors. They informed IDEA's prompts and the descriptors for one holistic rubric criterion per dimension; they were not independently scored as separate rubric items.

The terminology used in this article distinguishes the overarching construct from its operationalization. Critical thinking (CT) denotes the broader construct; critical thinking skills (CTS) denotes component cognitive operations; and subskill refers to each of the four dimensions selected for instruction and assessment. Enactment denotes observable evidence of a subskill in a written text, whereas performance denotes the corresponding rubric score. Development is reserved for claims about change in underlying capacity over time and is not inferred from this single-semester, uncontrolled sequence. Facione's framework also includes inference and self-regulation. Self-regulation was explicitly incorporated into IDEA's process through self-evaluation and revision, but it was not scored as a fifth textual outcome because the accuracy and quality of self-regulation cannot be inferred reliably from a final text alone. Inferential moves were treated as embedded within analysis, explanation, and evaluation rather than as a separate rubric dimension because

the writing tasks and assessment rubric were designed to capture inference as part of these broader reasoning operations, not as an independently elicited or scored outcome. The four selected subskills therefore reflect alignment among the course tasks, platform prompts, and rubric; they are not presented as the only or universally core components of CT.

This operationalization also has important boundaries. IDEA and the rubric share the same operational account of what critical thinking looks like in writing. Higher scores may therefore partly reflect closer alignment with the textual features made salient by the scaffold; the study cannot determine whether this alignment represents better reasoning beyond the adopted operational definition. Facione's framework also foregrounds individual cognitive operations and does not fully represent the social, institutional, and political dimensions through which disciplinary communities determine what counts as warranted reasoning. In these teacher-education tasks, judgments were situated within disciplinary and institutional norms. The scores should therefore be read as performance under one theory-informed framework, not as a neutral measure of critical thinking as a whole.

### 2.2. Critical Thinking in Teacher Education

In teacher education, CT is an important component of professional preparation, as future educators are expected to question pedagogical assumptions, make reasoned instructional decisions, and respond reflectively to complex classroom situations (Barahona et al., 2023; Lithoxoidou & Papadopoulou, 2024). Professional teaching also requires educators to integrate critical-thinking, disciplinary, and pedagogical knowledge when making instructional decisions and creating opportunities for students to reason critically about subject matter (Leibovitch et al., 2025).

Despite broad agreement about its importance, CT is not always made explicit in teacher-education curricula and learning tasks. When CT is treated as a generic, transferable skill with limited connection to disciplinary knowledge or pedagogical practice, what it requires in authentic professional situations can remain underspecified (Wason, 2025; Leibovitch et al., 2025). Pre-service teachers may therefore struggle to determine what CT requires in concrete academic and professional tasks, leading to superficial or procedural engagement with its demands (Fitriani & Prodjosantoso, 2024; Lithoxoidou & Papadopoulou, 2024). Research in teacher education has accordingly emphasized the value of structured and situated learning activities that make critical-thinking demands explicit and connect them with pedagogical practice (Barahona et al., 2023; Wason, 2025).

One way to address this challenge is to treat CT as a teachable, discipline-embedded construct enacted through discipline- and profession-relevant tasks. Analytical and argumentative writing provides a particularly useful site for this purpose because students must make reasoning visible through claims, evidence, justification, criteria, and evaluative judgments (Fraser et al., 2024; Wingate, 2012; Gomez-Laich et al., 2019). Academic writing was therefore selected as a domain-specific site for examining CT enactment, not as a proxy for general CT ability or overall writing quality. Without explicit support, however, students may not consistently make the reasoning demanded by such tasks visible in their written responses (Fraser et al., 2024; Yuan et al., 2020). In teacher education, these writing demands are additionally situated within professional reasoning, as students must justify how disciplinary content should be taught, adapted, and evaluated in school contexts (Barahona et al., 2023; Lithoxoidou & Papadopoulou, 2024). This combination of disciplinary and pedagogical reasoning makes teacher-education writing a particularly relevant context for explicit scaffolding.

### 2.3. Technology-Mediated Scaffolding for Critical Thinking in Writing

Technology-mediated scaffolding can support critical thinking in academic writing by externalizing complex cognitive operations and making them visible, teachable, and revisable during composition. From a sociocognitive perspective, digital scaffolds distribute support across prompts, guiding questions, and opportunities for iterative revision (Strobl et al., 2019). Research on CT instruction suggests that explicit teaching can produce stronger gains than incidental exposure when learners receive clear criteria, structured practice, and feedback (Marin & Halpern, 2011; El Soufi & See, 2019).

Academic and professional genres such as reports, case analyses, and argumentative essays demand more than textual fluency; they require disciplinary reasoning, including weighing evidence, positioning claims, and anticipating counterarguments (Wingate, 2012). Writers realize this reasoning through specific discursive choices, such as defining problems, comparing perspectives, evaluating evidence, and qualifying conclusions (Schleppegrell, 2001; J. Li & Mak, 2022). The central affordance of technology-mediated scaffolding is its capacity to make these demands visible and actionable, thereby translating critical thinking from an implicit expectation into a set of situated writing moves.

Existing tools realize this potential unevenly. Automated writing evaluation systems and grammar checkers typically evaluate surface form, including spelling, grammar, punctuation, and lexical repetition; some also address organization or cohesion, but they generally do not establish whether disciplinary reasoning is sound (Strobl et al., 2019). Peer-review platforms can structure feedback dialogue (M. Li, 2018), while argument-mapping tools can make relationships among claims, evidence, and counterarguments visible (W. M. Davies, 2012; Twardy, 2004). These approaches provide complementary forms of support, but few tools integrate planning, drafting, self-evaluation, feedback, and revision within a single workflow while keeping the source and function of feedback transparent (Strobl et al., 2019; Rapp & Kauf, 2018). IDEA was designed within this space while keeping formal and CTS-oriented feedback functions separate. Its real-time checks addressed only formal features such as sentence-final punctuation and word repetition, whereas its CTS-aligned messages were prewritten and triggered by students' checklist selections. The platform did not evaluate semantic accuracy, argument strength, or overall language quality.

This distinction has become more important amid growing use of generative AI for writing. These tools can readily provide fluent prose, reformulations, and feedback (Bittle & El-Gayar, 2025), but unstructured use may promote cognitive offloading (Gerlich, 2025; Roy & Jony, 2026), weaken learners' epistemic agency (Jose et al., 2025), or obscure whether the reasoning in a text originated with the student or the system (Tang, 2025). These risks should not be treated as inevitable consequences of all AI use. In a quasi-experimental study with pre-service preschool teachers, X. Zhang and Zhang (2026) examined an AI pedagogical agent with metacognitive scaffolding and modeled AI dependency as a moderator of the relationship between critical thinking and creative instructional-design performance. Their design illustrates how AI dependency can be examined alongside cognitive outcomes rather than inferred from tool use alone. These findings do not provide evidence about IDEA or resolve questions of textual provenance and epistemic agency in the present study. IDEA offers a contrasting, structured, non-generative design: it neither composes nor rewrites students' text, and it does not use AI to assess the semantic quality of their arguments. Instead, IDEA prompts students to select and apply targeted critical-thinking operations and to review and revise their own writing. The platform itself does not generate or reformulate students' text, although its prompts, descriptors, and feedback criteria structure the reasoning process in advance.

## 3. Materials and Methods

### 3.1. Research Design

This study used a design-based research (DBR) approach to develop, implement, and examine an initial technology-mediated scaffolding model for supporting critical thinking in academic writing. DBR was appropriate because the study investigated an educational innovation in an authentic learning context while generating practical design knowledge and theoretically informed insights for teaching critical thinking in teacher education (Barab & Squire, 2004; Anderson & Shattuck, 2012; Reeves & McKenney, 2012). The study focused on the situated enactment of interpretation, analysis, explanation, and evaluation in written academic tasks rather than on critical thinking as a general, abstract competence. This focus reflected the model's purpose: to make critical thinking visible, teachable, and assessable through writing.

The article reports the model's first complete DBR cycle: analysis of the instructional problem, formulation of design principles, implementation of corresponding platform features in two parallel courses, examination of performance and perception evidence, and specification of revisions for Cycle 2. The cycle was exploratory and intended to refine the design.

Within the DBR cycle, the empirical component followed a fixed-sequence repeated-measures design in which participants completed three writing tasks. Tasks 1 and 2 were completed without IDEA; Task 3 used the full scaffold. Consequently, platform use was confounded with time, task order, cumulative writing practice, increasing disciplinary knowledge, prompt differences, and scaffold–rubric alignment, because Task 3 was completed within an environment built around the same operational descriptors used in the analytical rubric. The repeated-measures ANOVAs test whether mean rubric scores differed across the task sequence; they cannot identify why those differences occurred.

No control or comparison group was included in this initial design-oriented cycle. The cycle prioritized implementation in an authentic instructional setting, ecological validity, instructional feasibility, and the generation of design principles for subsequent refinement and more controlled testing.

The design supported an integrated examination of the model across quantitative and qualitative evidence. Scores from the three writing tasks were used to analyze changes in critical-thinking performance across the fixed task sequence, while focus group accounts were used to examine how participants experienced IDEA, including perceived support and constraints in enacting the targeted subskills during writing.

### 3.2. Context and Participants

The study was conducted over one semester in two undergraduate courses on disciplinary grammar instruction at one highly selective Chilean university. The courses belonged to different teacher-education programs within the institution, but all participants were fourth-year pre-service teachers specializing in language education. Both were core courses in the professional curriculum, designed to connect conceptual knowledge of grammar with pedagogical knowledge for teaching grammar in schools.

> The two courses were treated as a single implementation cohort because they followed the same Cycle 1 protocol, including the same syllabus, instructor, assessment rubric, task sequence, and instructional conditions. The final analytic sample comprised 57 participants: Course A ($n = 27$) and Course B ($n = 30$). A post hoc exploratory empirical check of this pooling decision using the participant-level data was conducted. For each subskill, a mixed-design analysis examined Course (A vs. B) as a between-participants factor and Task (T1, T2, T3) as a

within-participants factor. Course-level descriptive statistics and the resulting Course × Task tests are reported in Supplementary Materials S1.

Both courses were taught by the same teacher educator and followed a common syllabus, similar assignments, and comparable pacing, which strengthened instructional consistency but also created potential bias because the lead researcher was both the instructor and the model's designer. Several procedures were used to mitigate this risk. Students' texts were assessed with an analytical rubric reviewed by external specialists and supported by an explicit scoring manual. Independent coders double-coded the entire corpus using anonymized folio identifiers rather than task labels, so they were not informed whether a text corresponded to Task 1, Task 2, or Task 3. Disagreements were resolved through discussion until consensus was reached. In addition, a research assistant moderated the focus group without the instructor present, and the transcripts were anonymized before analysis. These measures were intended to reduce the influence of instructor expectations on the scoring of students' writing and the collection of their perceptions.

The study included 57 fourth-year pre-service teachers specializing in language education. Their courses formed part of a professional disciplinary component intended to develop conceptual knowledge of grammar and pedagogical knowledge for grammar instruction. Although the participants had previous academic writing experience, prior to this course they had not received explicit instruction in the targeted critical-thinking subskills or in how to enact them in professional writing about language teaching. Institutional ethics approval was obtained before data collection, and all participants provided informed consent. Participation did not alter regular course requirements; the analyzed writing tasks were part of the semester's normal instructional sequence.

### Semester Timeline and Task Sequence

Both courses followed the same semester sequence shown in Table 2. Critical-thinking instruction began at the start of the course. Accordingly, Task 1 represents an early-semester measurement within the CT instructional sequence rather than a preinstruction baseline. IDEA was introduced only after Task 2, through an orientation conducted before Task 3. Because the tasks were embedded in regular coursework rather than scheduled as experimental sessions, the retained analytic records identify early-, mid-, and late-semester administration but do not support reporting exact elapsed weeks. The writing tasks were completed during scheduled classroom sessions. Students were instructed not to use external generative-AI tools while completing the tasks, and no instances of such use were observed or recorded during their administration.

**Table 2.** Semester sequence, instructional exposure, and task conditions.

| Relative Timing | Course Event | CTS Instruction | IDEA Status | Measurement/Design Note |
|---|---|---|---|---|
| Early semester | Task 1 | CTS purpose introduced; no platform-guided enactment | Not used | Four subskills scored with the common 1–4 rubric. |
| Between Tasks 1 and 2 | Regular coursework | Interpretation, analysis, explanation, and evaluation taught as reasoning and writing actions | Not used | Continued disciplinary instruction and writing practice. |
| Midsemester | Task 2 | Regular CTS instruction continued | Not used | Same four scored dimensions; prompt content differed from Task 1. |

**Table 2.** *Cont.*

| Relative Timing | Course Event | CTS Instruction | IDEA Status | Measurement/Design Note |
|---|---|---|---|---|
| After Task 2 | IDEA orientation | Four-stage workflow demonstrated before Task 3 | Introduced | Orientation occurred only after the second unscaffolded task. |
| Late semester | Task 3 | Regular instruction plus platform prompts, checklist, and feedback | Required | Same rubric; IDEA used only for Task 3. See Section 3.1 for the competing explanations of score change, including scaffold–rubric alignment. |
| End of semester | Focus group | Not an instructional event | Experience discussed | Twelve volunteers; perceived support and constraints examined. |

### *3.3. The Scaffolding Model*

The writing support model was designed to support the enactment of the four targeted critical-thinking subskills in academic writing. Drawing on the operationalization established in Section 2.1, the model incorporated interpretation, analysis, explanation, and evaluation into the writing process. Rather than assuming that learners could translate these subskill labels into written reasoning, IDEA made them actionable through subskill-aligned prompts, descriptive guidance, and self-evaluation criteria (see Supplementary Materials S2). This design was intended to make the targeted reasoning operations explicit within concrete, discipline-relevant writing activities, consistent with research supporting explicit CT instruction and scaffolded practice (Marin & Halpern, 2011; El Soufi & See, 2019; Fan & Chen, 2021).

IDEA (Interfaz de Escritura Andamiada/Scaffolded Writing Interface) was the technological vehicle for the model. It was developed collaboratively through regular design meetings, testing, and incremental technical revision before classroom implementation. Students received a task-specific orientation after Task 2 and before using IDEA for Task 3. The platform was organized into four pedagogical stages: skill selection and a guiding question, guided writing, self-evaluation, and prewritten feedback with final revision.

#### 3.3.1. Pedagogical Stages of the Platform

Stage 1—Skill Selection and Guiding Question: On entering the platform, students selected one to four critical-thinking subskills to target in the task: interpretation, analysis, explanation, and/or evaluation. They then entered the guiding question or topic for their text and highlighted its key terms in the interface. This stage was intended to help students identify the task focus and make salient the key terms that would organize their response.

Stage 2—Writing: Students worked through the prompts associated with their selected subskills. For each selected subskill, the platform offered structured prompts, descriptive guidance, and categorized discourse connectors to support lexical choice and cohesion. Students could compose directly in the platform or upload a draft, while retaining working notes in the same environment. A real-time function provided rule-based form feedback, flagging issues such as missing sentence-final punctuation or excessive word repetition. These checks addressed predefined formal features, while the structured prompts addressed CTS-aligned reasoning and argumentative organization (Strobl et al., 2019). The form feedback relied on predefined rules rather than generative AI or open-ended natural-language generation.

Stage 3—Self-Evaluation: After drafting, students viewed the full text and completed a checklist aligned with the subskills they had selected. The checklist was intended to support

reflection (Cheng & Chan, 2019), but in IDEA it functioned as a self-report mechanism: students could choose any response, including a response intended to elicit a particular feedback message. IDEA did not verify the accuracy of these selections, and the study did not compare checklist responses with independent rubric scores. The checklist therefore cannot be treated as a measure of self-evaluation accuracy.

Stage 4—Prewritten Feedback and Final Revision: Checklist selections triggered stored feedback messages aligned with the four subskills. This adaptation was limited to matching each selected checklist response with a predefined message. The platform did not interpret the semantic quality of the text, evaluate argument strength, or generate language with AI. Students could revise and submit their text, which IDEA stored for the study. The unverified self-report trigger is a design limitation and a priority for Cycle 2 data capture.

Figure 1 summarizes IDEA's workflow and feedback logic. It displays final revision and submission as a separate workflow step and distinguishes rule-based form feedback during drafting from prewritten CTS-aligned feedback triggered by students' checklist selections.

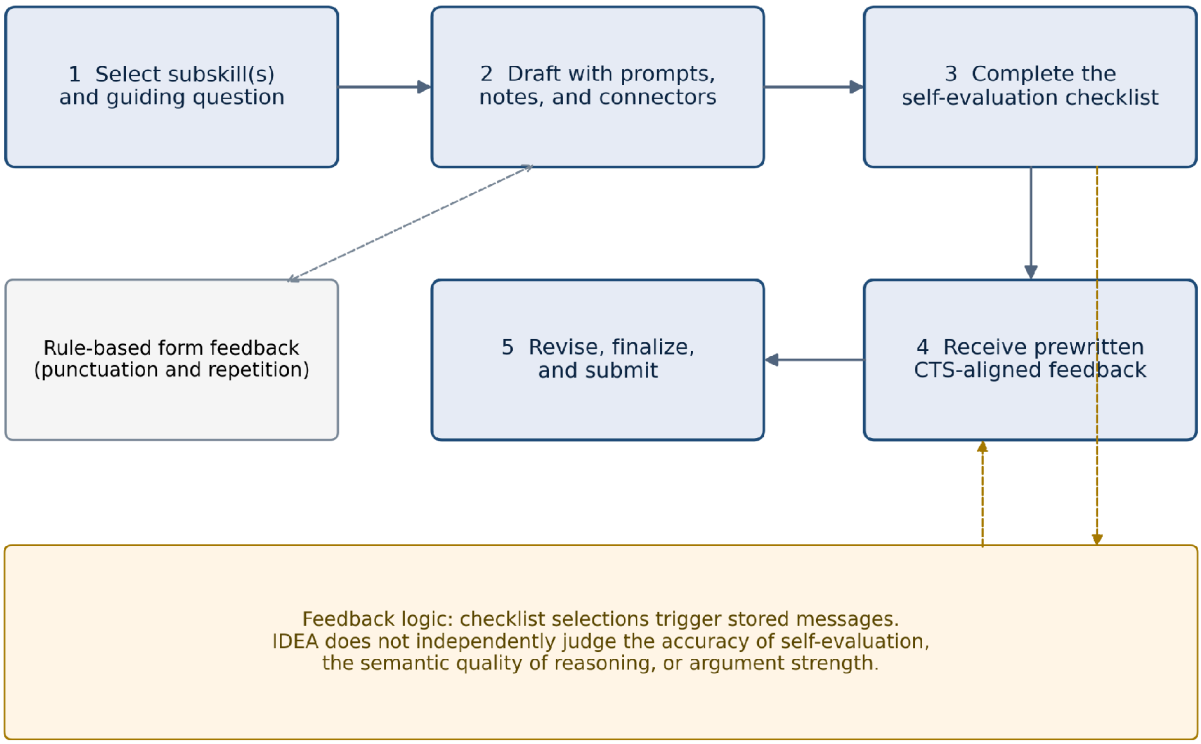



**Figure 1.** IDEA workflow and feedback logic. Solid arrows indicate the main sequence of the writing workflow; dashed arrows indicate feedback-related connections that operate alongside the main sequence.

IDEA was designed to address three documented limitations of digital writing support. First, many automated writing evaluation tools emphasize product-oriented feedback on formal features such as spelling and grammar, while offering more limited support for the composing process itself (Stevenson, 2016; Stevenson & Phakiti, 2014). Second, explicit support for higher-order reasoning is not consistently integrated into digital writing-support environments (Strobl et al., 2019; Yin et al., 2024). Third, writing support can be fragmented across different stages of composing, motivating designs that integrate planning, drafting, feedback, and revision within a coherent workflow (Strobl et al., 2019; Rapp & Kauf, 2018). IDEA therefore combines process-oriented scaffolding, explicit support for the targeted critical-thinking subskills, rule-based form feedback, and checklist-triggered prewritten CTS-aligned feedback within one environment.

The design also reflected principles of technology-mediated scaffolding relevant to academic writing: making task demands explicit through prompts and guidance, supporting self-evaluation and revision, and integrating these forms of support within the disciplinary writing process (Strobl et al., 2019; Rapp & Kauf, 2018). Consistent with research on explicit CT instruction, IDEA treated the four targeted subskills as teachable reasoning operations embedded in discipline-relevant writing tasks (Marin & Halpern, 2011; El Soufi & See, 2019).

#### 3.3.2. DBR Cycle 1: Conjectures and Revisions

Cycle 1 was organized around explicit design conjectures linking instructional problems, design principles, implemented features, evidence, and subsequent revision. Table 3 summarizes this conjecture–revision logic. The evidence column identifies the data or design limitations used to examine each conjecture; it should not be read as causal confirmation of the proposed mechanism. The Cycle 2 revisions are design outputs of the present cycle and were not part of the Cycle 1 implementation.

**Table 3.** Design conjecture–revision matrix for DBR Cycle 1.

| Instructional Problem | Design Principle | IDEA Feature | Evidence Examined in Cycle 1 | Revision for Cycle 2 |
| --- | --- | --- | --- | --- |
| CTS task demands can remain underspecified | Make reasoning operations explicit | Definitions, prompts, and guiding questions for four subskills | Participant reports that academic verbs and task demands became more concrete; rubric-score changes across tasks were examined descriptively alongside these accounts | Add contrasting examples and log which prompts students use |
| Writing support can be fragmented across composing stages | Integrate stages of composition | Selection, drafting, self-evaluation, feedback, and revision in one workflow | Participants described the integrated sequence as supporting the organization of ideas and the writing process | Retain the integrated workflow and add process logs to document use of its stages |
| The platform lacked longitudinal access to prior texts and progress | Support continuity across writing episodes | Texts could be submitted, but user accounts and longitudinal retrieval were not available | Participants requested user accounts, text retrieval, and progress access | Add user accounts, version history, and longitudinal progress views |
| Feedback is often surface-level or generic | Separate form feedback from reasoning support | Rule-based punctuation/repetition checks plus CTS-aligned stored messages | Participants reported insufficient revision guidance in some feedback, including rule-based feedback on word repetition | Provide more specific revision guidance and maintain separate logs for form feedback and CTS-aligned feedback |
| Self-evaluation accuracy is unverified | Retain learner-controlled selection while making the feedback trigger auditable | Student-selected checklist responses trigger stored messages | Checklist selections were not compared with independent rubric scores | Require text evidence for selections, log skill and checklist selections, and compare self-ratings with independent rubric scores |

**Table 3.** *Cont.*

| Instructional Problem | Design Principle | IDEA Feature | Evidence Examined in Cycle 1 | Revision for Cycle 2 |
|---|---|---|---|---|
| Change across tasks has competing explanations | Strengthen inference and task documentation | Three-task fixed-sequence repeated-measures design with a common rubric | Scores changed across tasks, but the design cannot separate IDEA use from the competing explanations described in Section 3.1, including scaffold–rubric alignment | Use a comparison condition, counterbalanced or independently equated prompts, fidelity logs, and exact timing records |
| Dimension-level measurement precision is not established | Strengthen task- and dimension-specific measurement evidence | Analytical rubric with four dimension scores | Per-dimension Cohen's kappa was computed from independent pre-consensus ratings; the pooled coefficients do not establish task-specific measurement precision | Collect task-specific inter-rater agreement evidence and examine whether the rubric functions comparably across tasks and course sections; report dimension- and task-specific agreement estimates with confidence intervals |

Taken together, Figure 1 and Table 3 make the design logic of IDEA explicit and inspectable. They document how the targeted critical-thinking subskills were translated into platform features, how those features were connected to specific instructional problems and design principles, what evidence was examined during Cycle 1, and how that evidence informed revisions for Cycle 2. This level of specification is intended to support scrutiny, adaptation, and reconstruction of the design by other research teams, while preserving the distinction between documenting a design and establishing its efficacy.

### *3.4. Instruments and Data Analysis*

Critical-thinking performance was the primary quantitative outcome. It was operationalized as rubric-scored textual evidence of interpretation, analysis, explanation, and evaluation in students' responses (Barahona et al., 2023; Rodríguez et al., 2022). The rubric did not produce a separate measure of overall writing quality, grammatical accuracy, fluency, or genre quality; any participant comments about those aspects are reported as perceptions rather than outcome results.

Data came from three structured academic writing tasks embedded in the regular courses. Supplementary Materials S3 reproduces the written instructions for all three tasks and describes their associated videos and children's work. The original videos and children's productions are not reproduced for privacy reasons. Consequently, readers can compare the written prompts and the described structure of the tasks but cannot independently judge the relative difficulty of the complete stimulus materials.

Although the tasks addressed different grammatical topics and were administered at different points in the semester, they were designed according to a common disciplinary-pedagogical structure. Across the three tasks, students worked within a common disciplinary-pedagogical framework focused on grammar teaching in school contexts. The grammatical content addressed in each task had been previously introduced and developed as part of the regular course, so participants approached the activities with prior familiarity with the relevant concepts. Each task used school-based instructional materials and began with a video created by other pre-service teachers showing either a pedagogical explanation or an actual teaching experience previously used with children to introduce

or explain grammatical content. Before responding to the writing prompts, participants watched the video and took notes on the disciplinary content presented, the examples and instructional resources used, and the way in which the concept was pedagogically explained. Students' responses were assessed using the same four-dimension analytical rubric and 1–4 performance scale.

Across the three tasks, students responded to four open-ended questions, each aligned with one of the four critical-thinking subskills assessed in the study: interpretation, analysis, explanation, and evaluation. The interpretation prompt required students to clarify the relevant disciplinary concept, identify the criteria or features used to characterize it, and apply that understanding to an example. The analysis prompt required them to examine how a pedagogical explanation was constructed and to relate instructional strategies or resources to the disciplinary content being taught. The explanation prompt required students to examine evidence of student performance or reasoning and explain the conceptual or reasoning difficulty reflected in it. Finally, the evaluation prompt required a criteria-based judgment of pedagogical explanations or resources, integrating disciplinary accuracy and pedagogical considerations.

Table 4 summarizes the documented task-specific characteristics across Tasks 1, 2, and 3, including disciplinary content, semester timing, IDEA use, instructional, application, and comparative materials, and the content-specific focus of the written responses. These characteristics are reported descriptively to make cross-task variation transparent. Although the tasks shared a common disciplinary-pedagogical structure, response format, targeted critical-thinking subskills, and analytical rubric, their relative difficulty was not independently assessed; accordingly, these shared features are treated as evidence of structural alignment rather than formal task equivalence.

**Table 4.** Task-specific characteristics across the three writing tasks.

| Criterion | Task 1 | Task 2 | Task 3 |
|---|---|---|---|
| Grammatical content | Adjectives | Pronouns | Verbs |
| Semester timing | Early semester | Midsemester | Late semester |
| IDEA use | Not used | Not used | Required |
| Primary instructional material | Video-based pedagogical explanation of adjectives | Video-based pedagogical explanation of pronouns | Video-based pedagogical explanation of verbs |
| Application material for explanation | Student work related to the identification and understanding of adjectives | Student work related to the identification and understanding of pronouns | Student work related to the identification and understanding of verbs |
| Comparative material for evaluation | Second pedagogical explanation of adjectives | Second pedagogical explanation of pronouns | Second pedagogical explanation of verbs |
| Content-specific focus of written responses | Disciplinary and pedagogical reasoning about the teaching of adjectives | Disciplinary and pedagogical reasoning about the teaching of pronouns | Disciplinary and pedagogical reasoning about the teaching of verbs |

As shown in Table 2, Tasks 1, 2, and 3 were administered in the early, middle, and late parts of the semester, respectively; exact week-to-week intervals could not be reconstructed from the retained records.

The tasks were structurally aligned, as described above and documented in Supplementary Materials S3, but their relative difficulty was not independently assessed. Structural alignment included the same disciplinary-pedagogical context, four-prompt organization, targeted critical-thinking subskills, response format, analytical rubric, and general types of instructional and application materials. The grammatical topic, semester

timing, and IDEA condition differed across tasks. Accordingly, the manuscript does not treat structural alignment as evidence of formal task equivalence or use it to exclude task-related confounding.

Students' texts were assessed with an analytical rubric designed to measure critical-thinking performance (Supplementary Materials S4). It contained four dimensions. Each dimension comprised one holistic criterion—interpretation, analysis, explanation, or evaluation—scored at one of four performance levels (1–4). Higher scores indicated more explicit, coherent, and contextually appropriate enactment of the corresponding subskill. Thus, there was one scored criterion per dimension, not multiple separately scored actions. The illustrative actions in Table 1 informed descriptor wording and scorer interpretation. Two external specialists reviewed the rubric before data collection: one with expertise in teacher education reviewed the relevance and coherence of the criteria in relation to pre-service teacher preparation, and one with expertise in educational linguistics and academic writing reviewed the clarity, precision, and disciplinary appropriateness of the descriptors. This expert review provides limited evidence regarding the content relevance, coherence, and clarity of the rubric; the instrument is not presented as a fully validated measure.

A multiphase scoring procedure was used to strengthen scoring reliability. First, a calibration phase was conducted using 50% of the written corpus, allowing the coders to refine their interpretation of the rubric descriptors, calibrate scoring criteria, and identify potential ambiguities in their application. This phase was guided by the application manual developed during the expert-review and rubric-refinement process, which provided illustrative examples for each criterion to support consistent scoring decisions. Following this calibration stage, two independent coders double-coded the complete dataset (171 text-level observations, comprising 57 participants across three tasks), including re-coding the calibration texts. The texts were identified using anonymized folio codes rather than task labels, and the coders were not informed whether a text corresponded to Task 1, Task 2, or Task 3. Unweighted Cohen's kappa coefficients were calculated from these independent, pre-consensus ratings separately for each subskill: interpretation, $\kappa = 0.88$ (exact agreement: 91.8%); analysis, $\kappa = 0.79$ (86.5%); explanation, $\kappa = 0.84$ (90.1%); evaluation, $\kappa = 0.83$ (88.9%), indicating substantial to almost perfect inter-rater agreement across all dimensions (Supplementary Materials S5). Disagreements were then reviewed and discussed until consensus was reached, and the resulting consensus scores entered the quantitative analyses.

Cronbach's alpha was subsequently calculated across the four dimension consensus scores using the pooled set of 171 text-level observations (57 participants $\times$ 3 tasks). It was not calculated across tasks within each dimension. The resulting coefficient was alpha = 0.790. Because observations from the same participant were repeated across the three tasks, this coefficient is reported strictly as a descriptive indicator of score coherence across the four rubric criteria rather than as evidence of temporal consistency, construct validity, or interchangeability among the dimensions.

Changes in critical-thinking performance were examined quantitatively. Descriptive statistics were calculated for each subskill. Separate repeated-measures ANOVAs compared interpretation, analysis, explanation, and evaluation scores across Tasks 1, 2, and 3. Although each rubric dimension used a four-point ordinal scale, repeated-measures ANOVA was retained as the primary analysis to estimate task means, mean differences, confidence intervals, and within-participant effect sizes. This treatment was regarded as an approximation and was not used as the sole basis for inference. All 57 participants completed all three tasks, so no participants were lost to attrition, and no incomplete cases were excluded from these analyses. Mauchly's test was used to assess sphericity; Greenhouse–Geisser-corrected results were reported throughout as a conservative approach. The four primary omnibus

ANOVA tests were evaluated using a Bonferroni-adjusted familywise significance threshold of $\alpha = 0.0125$ (0.05/4). All four omnibus tests remained statistically significant under this criterion (all $p < 0.001$). Bonferroni-adjusted pairwise comparisons were conducted within each subskill. Effect sizes were reported as partial eta squared ($\eta_p^2$) and generalized eta squared ($\eta_G^2$). Partial eta squared is design-specific and can be inflated in within-participant analyses; generalized eta squared is additionally reported to facilitate more transparent interpretation of effects in this repeated-measures design. Neither statistic identifies the source of score change or should be interpreted as evidence of an independent IDEA effect.

As a post hoc exploratory check of the decision to pool the two course sections, separate 2 × 3 mixed-design ANOVAs were also estimated for each subskill, with Course (A, B) as the between-participants factor and Task (T1, T2, T3) as the within-participants factor. Greenhouse–Geisser-corrected results were used for effects involving Task. The Course × Task interaction was the primary quantity of interest because it tests whether score-change patterns across the three tasks differed between sections. The four interaction $p$-values were adjusted using the Holm procedure. These analyses were conducted to assess possible section-level heterogeneity in the observed task sequence and do not alter the non-causal interpretation of the primary repeated-measures analyses.

To assess the robustness of the conclusions to the ordinal nature of the scores, non-parametric sensitivity analyses were also conducted. Friedman tests were performed separately for the four subskills, followed by paired Wilcoxon signed-rank tests for the three task comparisons within each subskill. The four Friedman-test $p$-values were adjusted using Bonferroni correction, with a familywise threshold of $\alpha = 0.0125$. Within each subskill, Wilcoxon post hoc $p$-values were Bonferroni-adjusted across the three pairwise comparisons. These sensitivity analyses treated the rubric scores as ordinal and were used to determine whether the pattern of statistical conclusions was consistent with that from the primary ANOVA analyses.

At semester end, 12 of the 57 students volunteered for an online focus group lasting approximately one hour. All were fourth-year pre-service language teachers enrolled in the two study courses. The focus-group records available for analysis did not contain a separate breakdown by age, gender, or program; consequently, subgroup characteristics and comparisons between volunteers and nonvolunteers cannot be reported. A research assistant moderated the session without the instructor present. The session was audio-recorded, transcribed verbatim, and anonymized before analysis (The focus group was conducted and transcribed in Spanish. Quotations reported in the manuscript were translated into English for publication and checked against the original Spanish transcript).

The qualitative analysis used a hybrid deductive-inductive codebook thematic approach with reliability-oriented calibration. It was not reflexive thematic analysis. Deductive codes represented scaffold use, the four subskills, perceived usefulness, and implementation challenges; inductive codes captured unanticipated accounts concerning self-regulation, metacognitive awareness, instructor feedback, and design needs. Themes were developed by grouping related codes and checking them against the complete transcript; they were not hierarchized as levels of importance.

For reporting, the semistructured guide can be summarized by five core prompts: (1) How did you use each stage of IDEA while completing Task 3? (2) What did the platform help you understand about interpretation, analysis, explanation, and evaluation? (3) In what ways, if any, did the prompts, checklist, or feedback influence how you planned, drafted, or revised? (4) What was difficult, unclear, or unhelpful? and (5) How, if at all, could a tool like IDEA affect feedback practices in teacher education? These are condensed English renderings of the guide's domains rather than a claim of verbatim English wording.

Two researchers independently coded an opening calibration segment and discussed discrepancies until they agreed on code meanings, scope, and boundaries. The lead researcher did not participate in the initial coding or codebook-calibration process. Both coders then applied the refined codebook to the remainder of the transcript, reviewed the complete set of coded extracts, and resolved disagreements through consensus. No numerical intercoder coefficient was used because agreement served codebook calibration and consensus interpretation. The lead researcher participated subsequently in the triangulation and interpretive review of the findings, after the coding framework and coded dataset had been established. Reflexivity was addressed by separating the lead researcher from initial coding and codebook calibration, while explicitly recognizing her dual role as instructor and designer during the subsequent triangulation and interpretation stage. Additional measures included using a moderator who was not the instructor, anonymizing the transcript before analysis, retaining critical design and usability accounts alongside favorable comments, and treating favorable comments as perceptions rather than evidence of platform effects.

ATLAS.ti supported the organization, retrieval, and comparison of coded segments but did not perform the analysis. The researchers made decisions about code development, theme construction, and interpretation through iterative reading, comparison, discussion, and refinement. This process produced four pedagogical-experience themes—CTS explicitness, reasoning organization, perceived support for self-regulation and metacognitive awareness, and perceived instructor-feedback affordance—and two design/usability categories: feedback specificity and accounts and longitudinal access.

## 4. Results

### *4.1. Changes in Critical Thinking Performance Across Tasks*

Separate repeated-measures ANOVAs examined changes in interpretation, analysis, explanation, and evaluation scores across the three tasks. Table 5 presents descriptive statistics, and Figure 2 shows the means with 95% confidence-interval error bars.

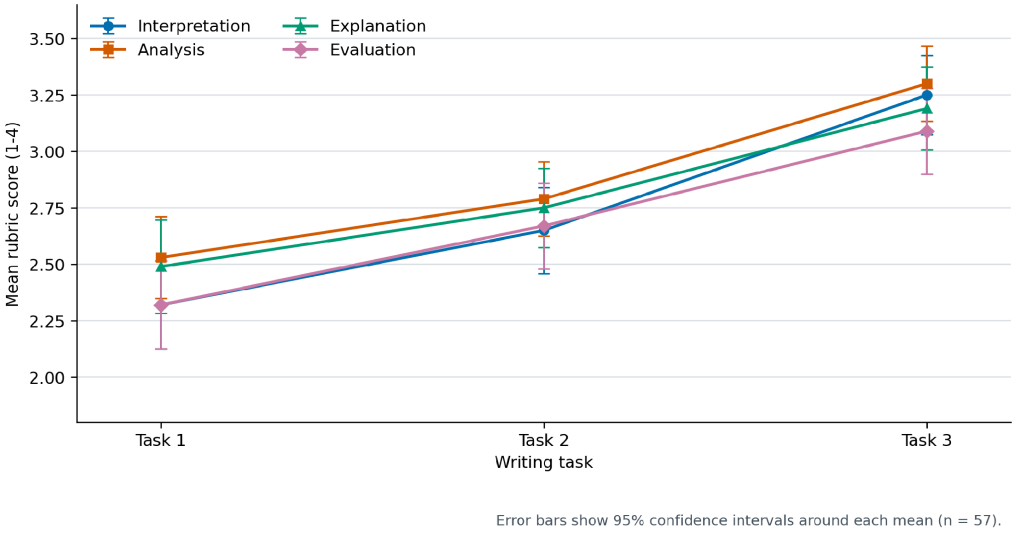



**Figure 2.** Critical–thinking performance across the three tasks.

**Table 5.** Mean critical-thinking scores by task.

| Critical Thinking Subskill | Task 1 | | Task 2 | | Task 3 | |
|---|---|---|---|---|---|---|
| | Mean | Standard Deviation | Mean | Standard Deviation | Mean | Standard Deviation |
| Interpretation | 2.32 | 0.736 | 2.65 | 0.719 | 3.25 | 0.662 |
| Analysis | 2.53 | 0.684 | 2.79 | 0.619 | 3.30 | 0.626 |
| Explanation | 2.49 | 0.782 | 2.75 | 0.662 | 3.19 | 0.693 |
| Evaluation | 2.32 | 0.736 | 2.67 | 0.715 | 3.09 | 0.714 |

Inferential analyses were conducted separately for the four subskills. Table 6 presents Mauchly's test, the Greenhouse–Geisser correction, corrected degrees of freedom, F and $p$ values, and both partial and generalized eta squared. Table 7 presents Bonferroni-adjusted pairwise comparisons. Positive mean differences in Table 7 are defined as the later task minus the earlier task. The ANOVA findings were supplemented by ordinal non-parametric sensitivity analyses reported below and in Supplementary Materials S6. These analyses describe change across the sequence; they do not isolate an IDEA effect or distinguish scaffold-related performance from closer alignment between the scaffold and the scoring rubric.

**Table 6.** Repeated-measures ANOVA results for the four critical-thinking subskills.

| Subskill | Mauchly's W | $p$ | GG ε | Corrected df | F | $p$ | Partial Eta Squared ($\eta_p^2$) | Generalized Eta Squared ($\eta_G^2$) |
|---|---|---|---|---|---|---|---|---|
| Interpretation | 0.937 | 0.169 | 0.941 | (1.88, 105.39) | 29.09 | <0.001 | 0.342 | 0.232 |
| Analysis | 0.904 | 0.063 | 0.912 | (1.82, 102.20) | 30.30 | <0.001 | 0.351 | 0.201 |
| Explanation | 0.903 | 0.060 | 0.911 | (1.82, 102.05) | 17.07 | <0.001 | 0.234 | 0.143 |
| Evaluation | 0.879 | 0.029 | 0.892 | (1.78, 99.94) | 16.55 | <0.001 | 0.228 | 0.163 |

Note. Greenhouse–Geisser-corrected degrees of freedom and $p$-values are reported for all subskills as a conservative approach. Mauchly's test rejected sphericity only for evaluation. The four primary omnibus ANOVA tests were evaluated using a Bonferroni familywise threshold of 0.0125; all four $p$-values were below 0.001. Partial eta squared and generalized eta squared are reported for the Task effect. Both effect sizes describe variance associated with task in this within-participant design and do not identify the cause of score change.

**Table 7.** Bonferroni-adjusted pairwise comparisons across tasks.

| Subskill | Comparison | ΔM (Later–Earlier) | 95% CI | Adjusted $p$ |
|---|---|---|---|---|
| Interpretation | Task 2–Task 1 | 0.33 | [0.06, 0.61] | 0.011 |
| Interpretation | Task 3–Task 2 | 0.60 | [0.30, 0.90] | <0.001 |
| Interpretation | Task 3–Task 1 | 0.93 | [0.59, 1.27] | <0.001 |
| Analysis | Task 2–Task 1 | 0.26 | [0.05, 0.48] | 0.013 |
| Analysis | Task 3–Task 2 | 0.51 | [0.27, 0.75] | <0.001 |
| Analysis | Task 3–Task 1 | 0.77 | [0.49, 1.06] | <0.001 |
| Explanation | Task 2–Task 1 | 0.26 | [0.01, 0.51] | 0.037 |
| Explanation | Task 3–Task 2 | 0.44 | [0.13, 0.75] | 0.003 |
| Explanation | Task 3–Task 1 | 0.70 | [0.37, 1.03] | <0.001 |
| Evaluation | Task 2–Task 1 | 0.35 | [−0.02, 0.72] | 0.067 |
| Evaluation | Task 3–Task 2 | 0.42 | [0.07, 0.77] | 0.013 |
| Evaluation | Task 3–Task 1 | 0.77 | [0.50, 1.04] | <0.001 |

Interpretation scores increased across tasks. Sphericity was met, W = 0.937, $p$ = 0.169. The task effect was statistically significant, F(1.88, 105.39) = 29.09, $p$ < 0.001, partial eta-squared = 0.342. Later-minus-earlier comparisons were: Task 2–Task 1, mean difference = 0.33, 95% CI [0.06, 0.61], adjusted $p$ = 0.011; Task 3–Task 2, 0.60 [0.30, 0.90], $p$ < 0.001; and Task 3–Task 1, 0.93 [0.59, 1.27], $p$ < 0.001.

Analysis scores increased across tasks. Sphericity was met, W = 0.904, $p = 0.063$. The task effect was statistically significant, F(1.82, 102.20) = 30.30, $p < 0.001$, partial eta-squared = 0.351. Later-minus-earlier comparisons were: Task 2–Task 1, mean difference = 0.26, 95% CI [0.05, 0.48], adjusted $p = 0.013$; Task 3–Task 2, 0.51 [0.27, 0.75], $p < 0.001$; and Task 3–Task 1, 0.77 [0.49, 1.06], $p < 0.001$.

Explanation scores increased across tasks. Sphericity was met, W = 0.903, $p = 0.060$. The task effect was statistically significant, F(1.82, 102.05) = 17.07, $p < 0.001$, partial eta-squared = 0.234. Later-minus-earlier comparisons were: Task 2–Task 1, mean difference = 0.26, 95% CI [0.01, 0.51], adjusted $p = 0.037$; Task 3–Task 2, 0.44 [0.13, 0.75], $p = 0.003$; and Task 3–Task 1, 0.70 [0.37, 1.03], $p < 0.001$.

Evaluation scores increased across tasks. Sphericity was violated, W = 0.879, $p = 0.029$, so the Greenhouse-Geisser correction was used. The task effect was statistically significant, $F(1.78, 99.94) = 16.55$, $p < 0.001$, partial eta-squared = 0.228. Later-minus-earlier comparisons were: Task 2–Task 1, mean difference = 0.35, 95% CI [−0.02, 0.72], adjusted $p = 0.067$; Task 3–Task 2, 0.42 [0.07, 0.77], $p = 0.013$; and Task 3–Task 1, 0.77 [0.50, 1.04], $p < 0.001$. Thus, the first interval was not statistically significant, whereas both comparisons involving Task 3 were significant.

Scores increased in all four subskills across the fixed task sequence. However, the reported pairwise tests compare task means; they do not test whether the Task 3–Task 2 change is significantly larger than the Task 2–Task 1 change. Accordingly, the adjacent-interval magnitudes are reported descriptively and are not treated as evidence that the pattern of score change differed at Task 3.

#### 4.1.1. Exploratory Check of Course-Level Heterogeneity

Because the pooled cohort comprised two intact course sections, we conducted a post hoc exploratory check of whether score patterns across tasks differed by course. At Task 1, Course A–Course B mean differences ranged from −0.18 to 0.33 points across the four subskills on the 1–4 scale. Course-specific descriptive statistics are provided in Supplementary Materials S1.

Mixed-design ANOVAs showed no statistically significant Course × Task interaction for interpretation, $F(1.89, 103.71) = 1.17$, $p = 0.312$, $\eta_p^2 = 0.021$; analysis, $F(1.84, 101.02) = 0.95$, $p = 0.384$, $\eta_p^2 = 0.017$; explanation, $F(1.83, 100.79) = 1.03$, $p = 0.354$, $\eta_p^2 = 0.018$; or evaluation, $F(1.78, 98.13) = 0.09$, $p = 0.899$, $\eta_p^2 = 0.002$. Holm adjustment across the four exploratory interaction tests left all conclusions unchanged (all adjusted $p = 1.000$). The main effect of Course was likewise not statistically significant for interpretation, analysis, explanation, or evaluation (all $p \geq 0.190$).

These results do not establish formal equivalence between sections. The exploratory analyses did not detect a statistically significant Course × Task interaction for any of the four subskills. Together with the shared instructional conditions and course-level descriptive results, this supports retaining the pooled cohort for the primary analyses. However, with 27 and 30 participants in the two sections, these tests cannot rule out smaller differences in score-change patterns. Course-level results are provided in Supplementary Materials S1 so that readers can examine this decision.

#### 4.1.2. Ordinal Sensitivity Analyses

Friedman tests, conducted as ordinal sensitivity analyses, confirmed differences across the three tasks for all four subskills: interpretation, $\chi^2(2) = 37.81$, $p < 0.001$, Kendall's W = 0.332; analysis, $\chi^2(2) = 39.05$, $p < 0.001$, Kendall's W = 0.343; explanation, $\chi^2(2) = 26.44$, $p < 0.001$, Kendall's W = 0.232; and evaluation, $\chi^2(2) = 28.22$, $p < 0.001$, Kendall's W = 0.248.

All four results remained statistically significant after Bonferroni adjustment across the four Friedman tests (Supplementary Materials S6).

Bonferroni-adjusted paired Wilcoxon signed-rank tests reproduced the same statistical conclusion as the ANOVA post hoc comparisons for each of the 12 contrasts: 11 were statistically significant and the Task 2–Task 1 comparison for evaluation was not (Wilcoxon $W = 546$, $p = 0.070$; Supplementary Materials S6). Thus, treating the four-point rubric scores explicitly as ordinal did not alter any of the significant/non-significant conclusions from the parametric analyses. This convergence supports the robustness of the reported score-change pattern across the fixed task sequence; it does not distinguish the competing explanations described in Section 3.1, including scaffold–rubric alignment.

### *4.2. Pre-Service Teachers' Perceptions of the Scaffolding Model*

The codebook analysis produced four pedagogical-experience themes—CTS explicitness, reasoning organization, perceived support for self-regulation and metacognitive awareness, and perceived instructor-feedback affordance—and two design/usability categories: feedback specificity and accounts and longitudinal access. Table 8 presents one representative excerpt for each theme or category. The excerpts illustrate the content of the themes, while participant counts indicate the number of unique focus-group participants whose coded contributions informed each theme or category. Each participant is counted only once per theme or category, regardless of the number of coded excerpts contributed. These counts are reported descriptively to show how broadly each theme or category was represented across the 12-person focus group and should not be interpreted as prevalence estimates beyond this volunteer sample or as measures of theme importance.

The distribution of contributions varied across themes and categories. CTS explicitness was represented across all 12 participants, followed by reasoning organization (11/12), self-regulation and metacognitive awareness (10/12), and perceived instructor-feedback affordance (6/12). Among the design/usability categories, feedback specificity was represented by eight participants, whereas accounts and longitudinal access was raised by five. These counts indicate the breadth of participant contributions to each theme or category within the focus group, rather than the frequency with which each topic was mentioned.

Participants reported that the scaffold made the meanings of analyze, interpret, explain, and evaluate more explicit by linking each verb to concrete writing actions. One participant described this problem directly: "In a test the question says 'analyse situation X or problem Y', but there is never a common understanding of what it means to analyse. Being given a step-by-step approach to a skill and linking it to how to express it is very helpful" (P6). These accounts concern participants' understanding of the scaffold, not a separate test of conceptual knowledge.

Participants also said that the prompts helped them organize paragraphs, clarify the function of sections, and make the logic of their reasoning more visible. These accounts concern perceived support for reasoning in writing; the study did not independently score overall organization or writing quality.

A third theme was perceived support for self-regulation and metacognitive awareness. Participants described the sequence of planning, drafting, self-evaluation, feedback, and revision as helping them examine and organize their writing process. The study did not observe or assess use outside Task 3; no claim of transfer or broader application is made from these accounts.

**Table 8.** Pedagogical-experience themes and design/usability categories.

| Theme/Category | Finding | Representative Evidence | Participants Contributing to Theme/Category ($n$ = 12) |
|---|---|---|---|
| Pedagogical experience—CTS explicitness | CTS became more explicit and teachable | "I used to think critical thinking was just 'thinking deeply,' but now I know what to look for and how to apply it" (P1). | 12 |
| Pedagogical experience—Reasoning organization | Reasoning was easier to organize and express | "It was easier to make sure each paragraph had a point, and that the point made sense" (P2). | 11 |
| Pedagogical experience—Self-regulation and metacognitive awareness | Perceived support for self-regulation and metacognitive awareness | "For me, the most significant thing was being able to make a rough draft with messy ideas. . . to look at while I was organizing the text" (P2). | 10 |
| Pedagogical experience—Perceived instructor-feedback affordance | More developed drafts might allow instructor feedback to focus on content and reasoning | "I think using resources like this allows the teacher to concentrate on telling me what is wrong with the content or my reasoning" (P1). | 6 ** |
| Design/usability—feedback specificity | Some form-feedback messages identified a problem without enough revision guidance | "Sometimes the platform feedback would tell me that I was repeating a word too many times. However, it did not tell me how to replace it" (P7). | 8 |
| Design/usability—Accounts and longitudinal access | Participants wanted logins and longitudinal access to texts | "I would like the platform to be able to have user logins. Sometimes I wanted to consult my texts or progress, and I had to save it on my computer" (P8). | 5 |

** The perceived instructor-feedback affordance was substantively developed by one participant. Five additional participants briefly expressed agreement with this idea, but did not elaborate on it or introduce additional dimensions. Accordingly, this theme was retained as a limited perceived affordance rather than a prominent pattern in the focus-group discussion.

One participant suggested that IDEA could help students submit more developed drafts and thereby allow instructor feedback to focus more on content and reasoning than on surface correction. No teacher-workload, feedback-content, or classroom-observation data were collected; this account is therefore a perceived affordance rather than evidence that pedagogical roles were redistributed.

Design and usability comments were analyzed separately. One category concerned feedback specificity: some rule-based form feedback identified a formal problem but did not provide enough revision guidance. One participant noted that rule-based feedback on word repetition identified a problem without providing a concrete revision strategy (P7). The second category concerned the absence of user accounts, storage, retrieval, and progress monitoring. Both categories directly informed Cycle 2 revisions. No participant account explicitly characterized the scaffold as making the writing process formulaic or compliance-oriented, or as constraining the reasoning process.

Overall, the focus group shows how 12 volunteers described IDEA's support and constraints. Their accounts help identify design features that were noticed, but they do not establish platform effects or represent all 57 participants. Because the volunteers discussed a platform designed by their instructor, favorable accounts are particularly vulnerable to expectancy and social-desirability pressures. The critical comments about feedback specificity and storage run against that pressure and are therefore especially informative for design refinement, although they do not make the sample representative.

## 5. Discussion

The findings comprise changes in rubric-scored critical-thinking performance across the three-task sequence and focus group volunteers' perceptions of IDEA. The discussion is organized around RQ1, RQ2, and the design conjectures generated for the next DBR cycle.

### *5.1. Research Question 1: Performance Across the Task Sequence*

Omnibus task effects were statistically significant for all four subskills, and 11 of the 12 adjusted pairwise comparisons were significant. Interpretation, analysis, and explanation increased significantly in both adjacent intervals; for evaluation, the first adjacent contrast was not significant. The reported pairwise tests do not establish that one adjacent interval was larger than the other. The evidence therefore supports upward change across the sequence, not a statistically demonstrated bend at Task 3. The higher scores observed at Task 3 should also be interpreted in light of scaffold–rubric alignment: IDEA made salient the same operational features of critical-thinking performance that were subsequently evaluated by the analytical rubric, so part of the observed increase may reflect closer alignment with those textual expectations rather than a broader change in underlying critical-thinking capacity. This pattern is consistent with literature favoring explicit instruction (Marin & Halpern, 2011; El Soufi & See, 2019), while explanations of the change remain hypotheses.

The four dimensions served as analytically distinct instructional and scoring categories while co-occurring in authentic texts. The pooled alpha indicates coherence among the four scores under this operationalization; it neither proves that they form a single interchangeable dimension nor establishes complete separability. Participants' reports that academic verbs became clearer are compatible with the design conjecture that decomposition makes tacit expectations more actionable (Facione, 1990; Thomas & Lok, 2015). This interpretation concerns alignment between design and participant accounts, not proof of a mechanism.

#### Precision and Variability in Evaluation Scores

The observed Task 2–Task 1 increase for evaluation (mean difference = 0.35) was comparable to, and slightly larger than, the corresponding increases for the other subskills. Its Bonferroni-adjusted 95% confidence interval was wider and crossed zero [−0.02, 0.72], whereas the interpretation interval was narrower [0.06, 0.61]. The non-significant evaluation contrast should therefore not be interpreted as evidence of a smaller increase or a different pattern of score change across the task sequence.

Evaluation scores showed greater uncertainty in this interval, and the omnibus analysis also required a Greenhouse–Geisser correction because sphericity was violated. The available results do not distinguish whether the wider uncertainty reflects heterogeneous participant performance, task characteristics, or measurement precision. Evaluation's pre-consensus Cohen's kappa was 0.83, within the range observed for the other subskills (0.79–0.88). However, these coefficients were calculated by pooling ratings across all three tasks within each subskill. They cannot show whether inter-rater agreement differed specifically between Tasks 1 and 2 or whether the rubric functioned comparably across tasks. Task-specific agreement evidence and further examination of whether the rubric functions comparably across tasks would be needed to assess whether measurement differences contributed to the wider interval.

### *5.2. Research Question 2: Participants' Perceptions of IDEA*

Focus group volunteers described writing as a way of making reasoning visible rather than only producing a formally correct text. Their accounts align with epistemic views of academic writing (Wingate, 2012; Schleppegrell, 2001), but they remain perceptions from 12 volunteers rather than independently observed changes in writing practice.

Participants valued the integrated sequence of selection, drafting, self-evaluation, feedback, and revision. They also described increased awareness of their own process. Because the study did not validate checklist accuracy or include an independent measure of metacognition, the appropriate conclusion is perceived support for self-regulation and metacognitive awareness, not demonstrated growth in autonomy.

As a non-generative tool, IDEA structured learners' activity without composing or reformulating their text. This preserves textual authorship and a visible locus of learner action, but textual authorship is not equivalent to epistemic agency. The prewritten prompts, descriptors, and feedback criteria structured reasoning in advance and were not set by the students. The study therefore shows that students produced and revised the submitted text; it did not measure their control over the epistemic criteria, goals, or judgments organizing that work. X. Zhang and Zhang (2026) illustrate one way to investigate this distinction empirically: their quasi-experimental study of metacognitive scaffolding in an AI pedagogical agent treated critical thinking as a mediator and AI dependency as a moderator. Although their study did not measure epistemic agency directly and does not provide evidence about IDEA, it shows how learner dependency and cognitive outcomes can be examined separately from textual authorship. No instances of external generative-AI use were observed.

Perceived Affordance for Instructor Feedback

One participant suggested that more developed drafts could allow instructors to devote more feedback to content, reasoning, and disciplinary understanding. This is a perceived affordance. The study collected no measures of teacher workload, feedback content, time allocation, or classroom interaction and therefore does not demonstrate a redistribution of pedagogical roles.

The available focus-group evidence does not establish subsequent classroom enactment, transfer, or a replicable pathway. These outcomes remain hypotheses for future study.

### *5.3. Design Conjectures and Revisions After Cycle 1*

A central contribution of Cycle 1 is the explicit articulation of the design logic underlying IDEA. The study documents how four critical-thinking subskills were operationalized as writing supports and connected to an integrated sequence of selection, drafting, self-evaluation, feedback, and revision. Figure 1 makes the implemented workflow visible, while Table 3 links instructional problems, design principles, platform features, evidence examined in Cycle 1, and concrete revisions for Cycle 2. Within this design logic, the strongest convergence concerned explicitness and process integration: participants reported that the resulting steps helped them understand task demands and organize reasoning. These findings support retaining and refining those design principles in Cycle 2. They do not establish that the same design will produce comparable outcomes in another course or institution.

Cycle 1 also exposed weaknesses in feedback specificity, data persistence, self-evaluation validity, task documentation, measurement precision, and causal inference. As summarized in Table 3, Cycle 2 should add more specific revision guidance, separate logs for form and CTS-aligned feedback, user accounts and version history, logged skill and checklist selections, comparisons between self-ratings and independent scores, exact timing records, stronger task-equivalence evidence, task-specific inter-rater agreement and examination of rubric comparability across tasks, and a comparison condition using counterbalancing or independently equated prompts.

### *5.4. Limitations and Interpretive Caution*

The absence of a control or comparison group precludes attributing score increases to IDEA. Platform use coincided with the final task and with additional time, practice,

content learning, and a different prompt. The three tasks were structurally aligned but were not independently equated for difficulty, so task-related differences remain a competing explanation for the observed score changes. In addition, because IDEA prompts, checklist criteria, and the scoring rubric shared one operational account of critical thinking in writing, higher Task 3 scores may partly reflect closer alignment with the textual features made salient by the scaffold. Partial eta-squared and generalized eta-squared values describe variance associated with task in this within-participant design; they do not identify the source of change and should not be compared directly with between-subjects benchmarks.

The study was also conducted in one institutional setting—a highly selective Chilean university—with fourth-year pre-service language teachers. The writing tasks were specific to pedagogical grammar courses and were assessed with a rubric developed and reviewed for this study. These features enhance ecological validity within the setting but limit generalizability. The findings may not extend to other institutions, disciplines, stages of preparation, or cultural contexts (Fitriani & Prodjosantoso, 2024; Benabbes et al., 2025).

The study represents one exploratory DBR cycle. Its principal contribution lies in the documented conjecture–revision logic presented in Table 3, which makes the relationships among instructional problems, design principles, implemented features, Cycle 1 evidence, and proposed Cycle 2 revisions explicit. Claims about stable efficacy and replicability require subsequent testing. Feedback specificity, accounts and storage, checklist validity, and data capture require revision before a stronger evaluation.

The focus group represents 12 volunteers and lacks a separate demographic profile. The lead researcher was the instructor and designer. Moderator independence, anonymization, consensus coding, the inclusion of critical design accounts, and cautious interpretation reduced, but did not remove, expectancy and social-desirability risks. Finally, transfer beyond Task 3 was not assessed.

### *5.5. Implications and Future Research Directions*

The findings justify further testing of digital designs that make reasoning operations explicit within authentic writing tasks. IDEA illustrates one candidate design that aligns prompts, self-evaluation, feedback, and a rubric; the present cycle does not establish a replicable pathway or a separate improvement in overall writing quality.

Future studies should use comparison groups and counterbalanced or independently equated prompts, preserve exact timing records, and report fidelity and external-AI use. Longitudinal work should test rather than assume transfer to other genres, disciplines, or professional settings. Independent measures of self-regulation and metacognition should complement self-report, and process logs should be compared with rubric scores. More specifically, a subsequent DBR cycle could compare IDEA's non-generative scaffold with an AI pedagogical agent that provides matched metacognitive prompts, while counterbalancing task order or staggering scaffold introduction across comparable courses. Following X. Zhang and Zhang (2026), AI dependency should be measured and tested as a moderator rather than being treated only as a background concern. Independent measures of critical thinking and epistemic agency, together with process logs, could then test whether any difference is attributable to the non-generative format, to the metacognitive scaffolding shared by both conditions, or to participants' dependency on AI. This comparison would address the offloading question empirically instead of inferring learner agency from textual authorship.

Later DBR cycles should examine more diverse institutions, disciplines, and levels of writing experience. The next cycle should first implement the feedback-specificity, separate feedback-log, user-account, version-history, and data-capture revisions specified in Table 3. AI-based feedback on higher-order reasoning may subsequently be explored,

but it should remain distinct from generative text production and be evaluated with explicit learner-agency safeguards.

**Supplementary Materials:** The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/educsci16101658/s1. The Supplementary Materials include Supplementary Materials S1: Course-Section Descriptive Statistics and Exploratory Mixed-Design Analyses; Supplementary Materials S2: IDEA Interface and Scaffolded Writing Model; Supplementary Materials S3: Writing Tasks: Critical Thinking Skills in a Disciplinary-Pedagogical Context; Supplementary Materials S4: Analytical Rubric for Assessing Critical Thinking Subskills in Written Responses; Supplementary Materials S5: Inter-Rater Agreement by Critical-Thinking Subskill; and Supplementary Materials S6: Parametric and Ordinal Sensitivity Analyses.

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## Abbreviation

The following abbreviation is used in this manuscript:

CTS Critical Thinking Skills

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